# PROJECT PROPOSAL

## Project Title

**Analysis and Recommendation of Beauty and Cosmetics Products Using Consumer Ratings, Product Features and Usage Patterns**

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# 1. Project Overview

The beauty and cosmetics industry contains a wide variety of products from different brands, categories, price ranges, ingredients, and suitability for different skin types. With the increasing number of products available, consumers may find it difficult to identify products that best suit their needs.

This project aims to analyze a dataset of beauty and cosmetics products to identify patterns in product prices, ratings, customer reviews, usage frequency, skin types, ingredients, brands, and other product characteristics.

The project will use **Exploratory Data Analysis (EDA)** and **machine learning techniques** to understand consumer and product patterns and develop a model that can help identify or recommend suitable beauty and cosmetic products.

The dataset contains **15,000 records and 14 variables**, including:

- Product Name
- Brand
- Category
- Usage Frequency
- Price
- Rating
- Number of Reviews
- Product Size
- Skin Type
- Gender Target
- Packaging Type
- Main Ingredient
- Cruelty-Free status
- Country of Origin

The final system is intended to provide useful insights for both consumers and businesses in the beauty and cosmetics industry.

# 2. Business Understanding

## 2.1 Business Problem

Consumers have many beauty and cosmetic products to choose from, but selecting an appropriate product can be difficult because products differ in price, ratings, ingredients, skin-type suitability, usage frequency, and other characteristics.

At the same time, beauty businesses need to understand which types of products are popular, highly rated, and likely to attract customers.

Therefore, there is a need for a data-driven system that can analyze product information and identify important patterns that can support better product selection and business decision-making.

## 2.2 Project Aim

The main aim of this project is to **analyze beauty and cosmetics product data and develop a data-driven product recommendation or prediction system that can help consumers identify suitable products while providing useful business insights.**

## 2.3 Specific Objectives

The project will:

1. Analyze the characteristics of beauty and cosmetic products.
2. Identify the most popular product categories and brands.
3. Analyze the relationship between product prices, ratings, and number of reviews.
4. Determine which products are associated with different skin types.
5. Investigate the relationship between usage frequency and product characteristics.
6. Identify common ingredients used in highly rated products.
7. Analyze consumer preferences across different product categories.
8. Develop a machine learning model for product recommendation or prediction.
9. Evaluate the performance of the developed model.
10. Provide recommendations that can support consumers and beauty businesses.

## 2.4 Key Business Questions

The project will attempt to answer questions such as:

- Which beauty product categories are most common?

- Which brands have the highest-rated products?
- Does price influence product ratings?
- Which products receive the largest number of reviews?
- Which ingredients are most frequently used?
- Which products are suitable for different skin types?
- What usage frequencies are most common?
- Are cruelty-free products associated with different ratings or prices?
- Which product characteristics can be used to recommend suitable products?

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# 3. Data Understanding

## 3.1 Dataset Description

The dataset contains **15,000 beauty and cosmetics product records** with **14 variables**.

The variables include both numerical and categorical information.

### Numerical Variables

- **Price_USD** – price of the product in US dollars.
- **Rating** – customer rating of the product.
- **Number_of_Reviews** – number of reviews associated with the product.

### Categorical Variables

- Product_Name
- Brand
- Category
- Usage_Frequency
- Product_Size
- Skin_Type
- Gender_Target
- Packaging_Type
- Main_Ingredient
- Country_of_Origin

### Boolean Variable

- **Cruelty_Free** – indicates whether the product is cruelty-free.

## 3.2 Data Quality Assessment

The dataset will be checked for:

- Missing values
- Duplicate records
- Incorrect data types
- Outliers
- Inconsistent categories
- Unusual prices or ratings
- Invalid values

The data will then be cleaned and prepared before analysis and modeling.

## 3.3 Data Preparation

The following preprocessing activities will be performed:

1. Identify and handle missing values.
2. Remove duplicate records where necessary.
3. Check numerical variables for outliers.
4. Convert variables into appropriate data types.
5. Encode categorical variables for machine learning.
6. Scale numerical variables where required.
7. Select relevant features for modeling.

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# 4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis will be used to understand the structure and patterns within the dataset.

## 4.1 Univariate Analysis

Individual variables will be analyzed using graphs such as:

- Histograms
- Bar charts
- Boxplots
- Count plots

Examples include:

**Product Category Distribution**

A bar chart will be used to determine which cosmetic categories occur most frequently.

### Product Rating Distribution

A histogram will show how customer ratings are distributed across the products.

### Price Distribution

A histogram and boxplot will be used to examine product prices and identify possible outliers.

### Skin Type Distribution

A count plot will show the number of products associated with different skin types.

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## 4.2 Bivariate Analysis

Relationships between two variables will be investigated.

Examples include:

### Price vs Rating

A scatter plot will determine whether there is a relationship between product price and customer rating.

### Price vs Number of Reviews

This analysis will investigate whether expensive products receive more or fewer reviews.

### Category vs Rating

A boxplot can be used to compare ratings across different product categories.

### Skin Type vs Product Category

A grouped bar chart can show which categories are commonly associated with different skin types.

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## 4.3 Multivariate Analysis

Multiple variables will be analyzed together to identify more complex patterns.

Possible analyses include:

- Price, rating, and number of reviews
- Category, skin type, and rating
- Brand, price, and rating
- Usage frequency, price, and rating

A correlation heatmap can also be used to examine relationships among numerical variables.

## Expected EDA Findings

The analysis is expected to reveal:

- Popular cosmetic categories.
- Frequently occurring brands.
- Common price ranges.
- Highly rated products.
- Products with high numbers of reviews.
- Popular ingredients.
- Differences in products targeted at different skin types.
- Patterns between price and customer ratings.
- Patterns between product ratings and number of reviews.

These findings will guide the selection of features and modeling techniques.

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# 5. Modeling and Evaluation

## 5.1 Modeling Approach

Machine learning will be used to develop a system that can assist with beauty product recommendations.

A **content-based recommendation approach** is suitable because the dataset contains detailed product characteristics such as:

- Category
- Brand
- Skin Type
- Main Ingredient
- Usage Frequency

- Price
- Product Size
- Gender Target
- Cruelty-Free status

The system can compare the characteristics of products and identify products with similar features.

# 5.2 Feature Engineering

Relevant product characteristics will be transformed into numerical representations.

Categorical variables can be encoded using techniques such as:

- One-hot encoding
- Label encoding

Numerical features such as price and rating can be standardized where necessary.

For text-based product information, techniques such as **TF-IDF** may be considered.

# 5.3 Possible Models

The project can investigate models such as:

### 1. Cosine Similarity

Cosine similarity can measure how similar two products are based on their features.

For example, if a user likes a particular serum designed for sensitive skin containing Aloe Vera, the system can identify other products with similar characteristics.

### 2. K-Nearest Neighbors (KNN)

KNN can identify products that are closest to a selected product based on their features.

### 3. Clustering

Algorithms such as **K-Means clustering** can group similar beauty products into clusters based on their characteristics.

# 5.4 Model Evaluation

The recommendation system will be evaluated using appropriate recommendation metrics, such as:

- Precision
- Recall
- F1-score
- Precision@K
- Recall@K

Where applicable, model performance will be compared to determine which approach produces the most useful recommendations.

The model should ideally recommend products that are relevant to the user's selected product characteristics.

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# 6. Deployment

After developing and evaluating the model, the final recommendation system can be deployed as a simple user-friendly application.

A possible deployment architecture is:

**User Input → Data Processing → Recommendation Model → Recommended Products → Results**

For example, a user could provide:

- Preferred skin type
- Product category
- Preferred price range
- Ingredient preference
- Desired rating
- Cruelty-free preference

The system would then return products that best match those preferences.

## Possible Deployment Tools

The project can be deployed using:

- **Python**
- **Pandas**
- **NumPy**

- **Scikit-learn**
- **Matplotlib**
- **Seaborn**
- **Streamlit**
- **Jupyter Notebook**

Streamlit can be used to create a simple web interface where users enter their preferences and receive product recommendations.

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# 7. Conclusion

This project will demonstrate how data science and machine learning can be applied to the beauty and cosmetics industry.

By analyzing 15,000 cosmetic product records, the project will identify important patterns related to product categories, brands, prices, ratings, reviews, ingredients, skin types, and usage frequency.

Exploratory Data Analysis will provide insights into consumer and product patterns, while machine learning will be used to develop a recommendation approach based on product characteristics.

The resulting system can help consumers make more informed product choices and help businesses understand product trends and customer preferences.

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# 8. Recommendations

Based on the expected analysis, the following recommendations can be made:

### For Consumers

1. Consumers should consider **skin type and product ingredients**, rather than relying only on price.
2. Customer ratings and number of reviews can be considered together when comparing products.
3. Consumers should use the recommendation system to identify products matching their preferences.
4. Consumers interested in ethical products can filter recommendations based on cruelty-free status.

### For Beauty Businesses

1. Businesses should identify and invest in product categories with strong customer interest.
2. Brands should monitor customer ratings and reviews to improve product quality.
3. Businesses can use customer and product data to develop targeted product recommendations.
4. Businesses should analyze popular ingredients and skin-type preferences when developing new products.
5. Pricing strategies should be informed by customer demand and product performance rather than price alone.
6. Businesses can use recommendation systems to improve customer engagement and product discovery.

### For Future Development

The project can be improved by collecting additional information such as:

- Actual customer purchase history
- Customer age
- Customer location
- Individual customer preferences
- Customer reviews and review text
- Purchase frequency
- Product availability
- Customer feedback

With this additional information, the system could eventually be developed into a more advanced **hybrid recommendation system** combining content-based and collaborative filtering techniques.

# Overall Expected Outcome

The final project will produce:

**A cleaned cosmetics dataset → EDA and visualizations → machine learning/recommendation model → model evaluation → deployable product recommendation system → business recommendations.**